

EE4410 CYBER SECURITY OF POWER GRIDS

Assignment 1: URL Phishing and Malware Identification

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In this lab report, techniques for identifying malware and phishing attacks were explored. Public threat intelligence tools, such as VirusTotal and Cybercrime Tracker, were used to analyze file hashes and suspicious URLs. The aim was to demonstrate how malicious indicators can be detected and interpreted.

1 Malware Identification Using VirusTotal

This section presents an analysis of file hashes and several malicious URLs using VirusTotal.

1.1 File Hash Analysis

Five SHA256 hashes were analyzed using VirusTotal. Each hash yielded a detailed scan result showing the detection ratio and classification by multiple antivirus engines.

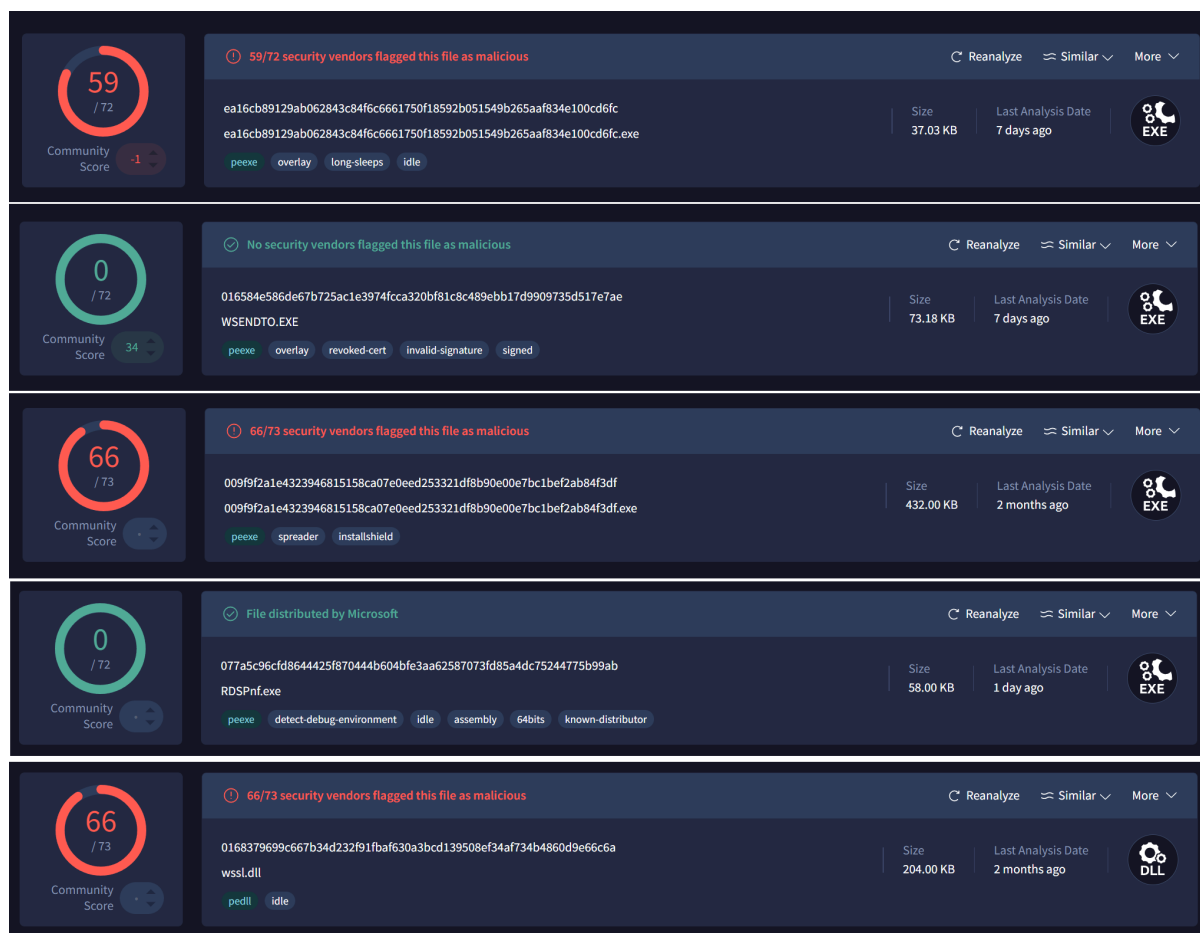


Figure 1: Virustotal scan result of the five hash signatures

Of the five analyzed file hashes, three were clearly identified as malicious by VirusTotal, as shown in Figure 1. Hence, three out of five samples were confirmed malware, while the other two were likely safe.

File signatures refer to the cryptographic hash of a file (in this case, SHA256), which acts as a unique fingerprint. These hashes allow antivirus tools to match the scanned file against known malicious

signatures in threat databases. A consistent match across engines increases the confidence in a file's classification as malware.

1.2 Malicious URLs

Three URLs were selected from the cybercrime-tracker.net platform and scanned using VirusTotal:

- <http://achillharpfestival.ie/wp-content/plugins/dbzytgojke/mail.php>
- 103.147.185.68/j/p30pq/login.php
- <http://unwaveringmedia.com/wp-content/plugins/vsqjsmplq/remain.php>

All three were identified as malicious by multiple security vendors and flagged for phishing, malware distribution, or suspicious behavior, as shown in Figure 2.

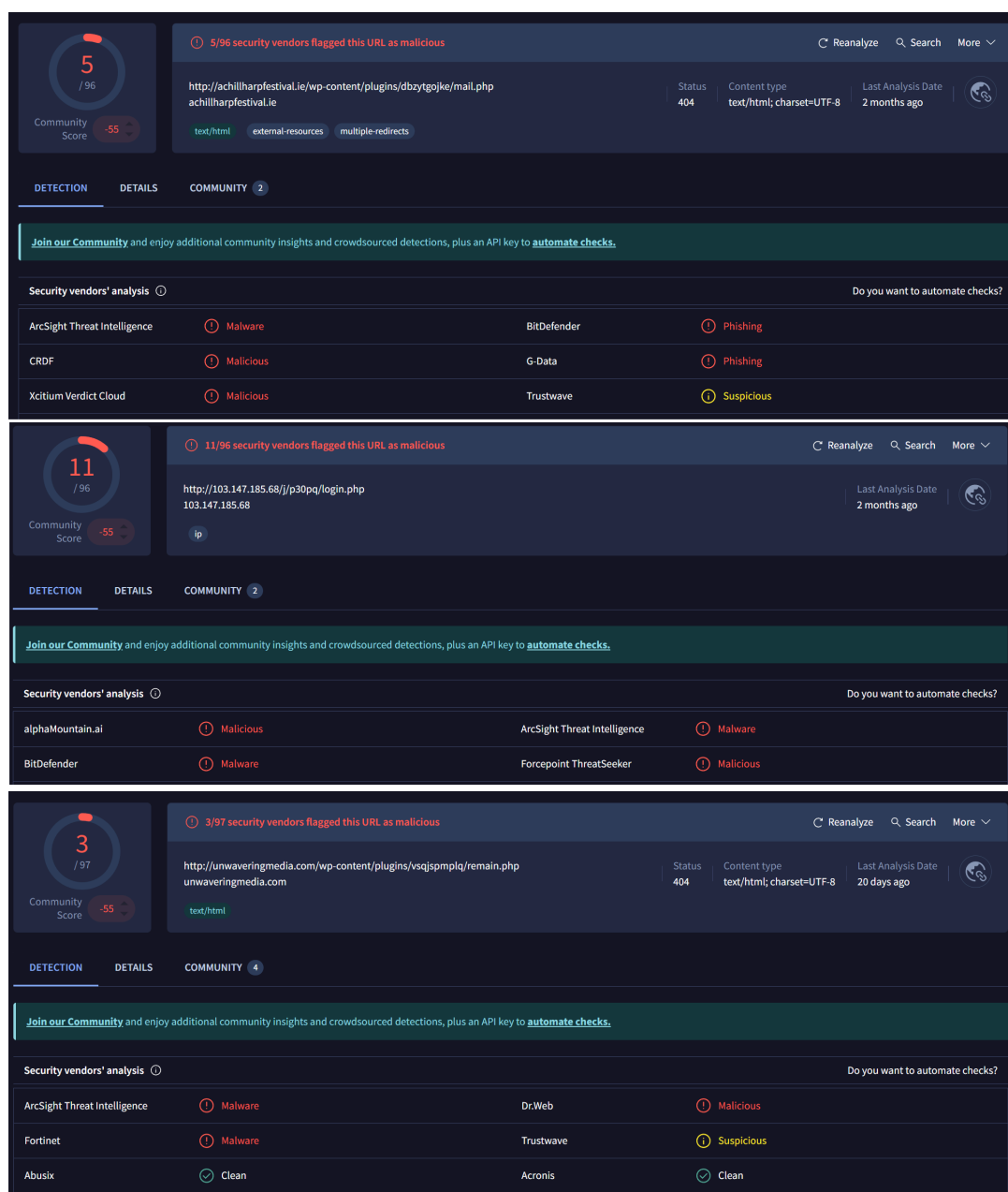


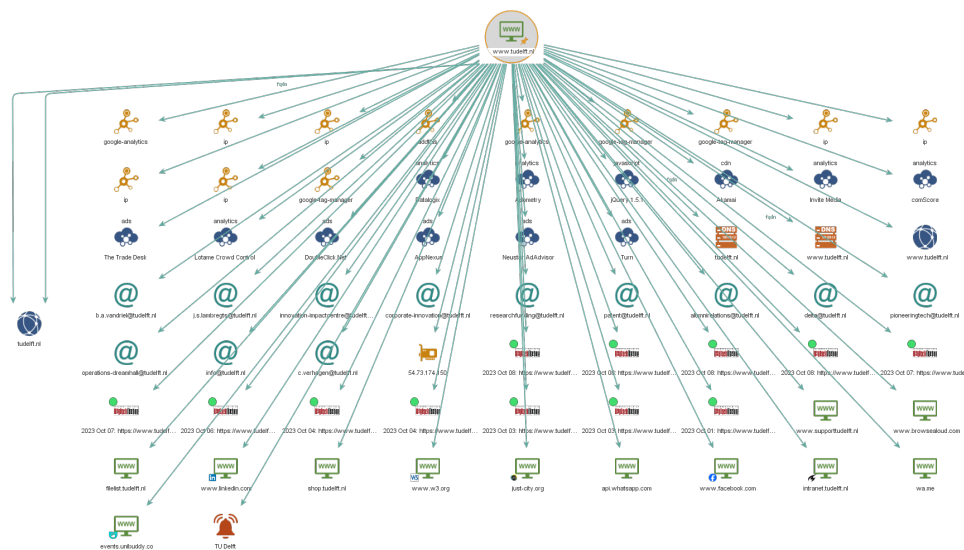
Figure 2: Virustotal scan result of three malicious URLs

The following common characteristics were observed among the malicious URLs:

- The analysis demonstrates how phishing sites disguise themselves to appear legitimate while performing background activities such as credential harvesting or malware deployment.

Maltego was used to perform Open-Source Intelligence (OSINT) on public web infrastructure and individuals. The objective was to simulate passive reconnaissance activities that an attacker might perform before launching a targeted cyber attack.

A new Maltego graph was initialized using the “Website” entity type. The domain `www.tudelft.nl` was entered, and all available transforms were executed, as shown in .



The results revealed detailed metadata, including:

- Email addresses associated with subdomains and services, such as `researchfunding@tudelft.nl`.
- Subdomains and hosting services
- Third-party integrations including Google Analytics, WhatsApp etc.
- DNS information such as IP addresses

2.2 Person Entity Investigation

It is also possible to perform a person search using Maltego by entering a name or alias and running available transforms. This reveals public data such as associated domains, usernames, and indexed online content,

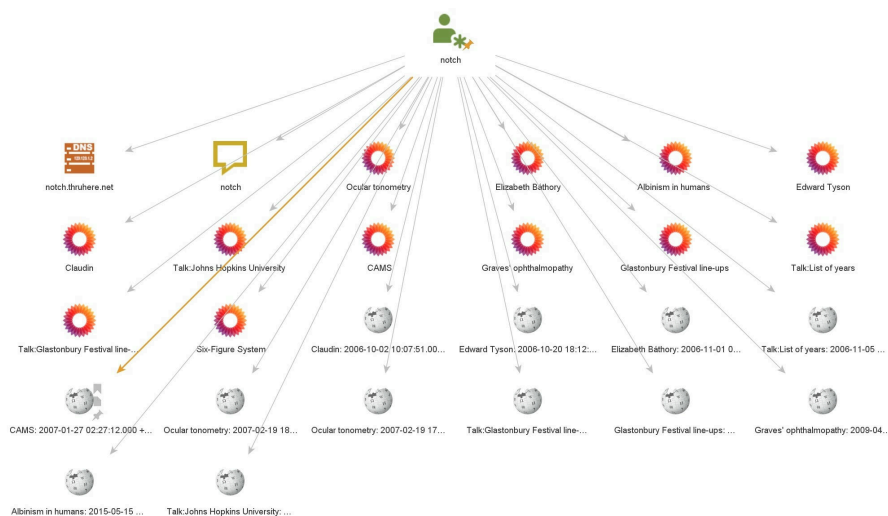


Figure 4: Example Maltego person graph

This technique can be used to identify potential impersonation, exposed infrastructure, or social activity linked to a target all without direct interaction.

2.3 Case Study: OSINT-Based Attack Scenario

In a simulated case, data extracted from the `www.tudelft.nl` domain showed email routing data pointing to third-party mail infrastructure. Public subdomains revealed backend services (`mail.tudelft.nl`, `vpn.tudelft.nl`) which, if unpatched, could be exploited further by attackers.

An adversary could:

1. Use DNS data to craft phishing emails spoofing institutional domains.
2. Search for exposed email credentials in known breach databases.
3. Launch password spraying attack or social engineering based on extracted employee names and positions.

This shows that OSINT is often the first step in reconnaissance, which enables attackers to launch spear phishing or credential-based attacks without breaching the perimeter.

2.4 Countermeasures

To reduce exposure and defend against OSINT-based threats:

- Limit publicly visible metadata (e.g., hide unnecessary DNS records)
- Use WHOIS privacy and remove personal information from public registries
- Regularly monitor breach databases for leaked credentials
- Train employees to minimize exposure on public social media